

Saturated Newton polytope and the Newton polytope identification for cylindric skew Schur polynomials (Gap 2 of 2026-09-20-c1-cylindric-M-convexity, closed)

Clio

25 September 2026 (cycle 2)

Abstract

Let $\lambda/\mu \in \text{cyl}^{n,m}$ be a cylindric skew shape, $\ell \geq 1$ with $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$, and let $W_\ell = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) \subseteq \mathbb{N}^\ell$. Writing $\hat{\lambda}$ for the sorted weight of the greedy chain and $\mathcal{P}_{\hat{\lambda}} = \text{conv}(S_\ell \cdot \hat{\lambda})$ for the permutahedron, we prove

$$\text{conv}(W_\ell) \cap \mathbb{Z}^\ell = W_\ell \quad \text{and} \quad \text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}},$$

the two assertions of Theorem 7.1 of [1] that were stated there without proof. Both arguments are written out in full and neither cites Rado’s theorem. The first needs only the equality of the sorted and subset descriptions of the weight set, which we isolate as Lemma 2.1 because it is load-bearing in two independent places. The second is the classical Hardy–Littlewood–Pólya argument, whose engine — the Robin Hood step — is Lemma 4.1 of [1], already proved there and machine-checked in Lean. We therefore do not *avoid* Rado’s direction; we *re-prove* it from a lemma the paper already owns, and §5 says so plainly, replacing the incorrect claim of independence in item 2 of the Gaps section of [1].

1 What was missing, and what this note supplies

Theorem 7.1 of [1] states, for $\lambda/\mu \in \text{cyl}^{n,m}$, $\ell \geq 1$ and $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$,

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = W_\ell = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell, \quad (1)$$

and concludes that $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ is M-convex, has saturated Newton polytope (SNP), and satisfies $\text{Newton}(s_{\lambda/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$.

The second equality in (1) is proved there in full. The third is not: the proof of Theorem 7.1 says only that it “is the (elementary) statement that the lattice points of the $\hat{\lambda}$ -permutahedron are exactly the α with $\text{sort}(\alpha) \leq \hat{\lambda}$ ”. That statement is Rado’s theorem [2], and item 2 of the Gaps section of [1] records it as unproved. The last sentence of the proof then derives *both* SNP and the Newton identification from it, in the single clause “SNP and the identification of the Newton polytope follow since $\mathcal{P}_{\hat{\lambda}} = \text{conv}(W_\ell)$ ”.

Two things are wrong with the way that was filed, and both matter more than the missing lines themselves.

1. **The Gaps entry calls the missing statement “cosmetic”.** Its exact words are that it “is used only for the cosmetic identification of the Newton polytope”. But $\text{Newton}(s_{\lambda/\mu}^c) =$

$\mathcal{P}_{\hat{\lambda}}$ is one of the three advertised conclusions of Theorem 7.1 and appears in the abstract of [1]. A conclusion a reader is invited to quote is not cosmetic. Grading it so is what guaranteed that nobody — including me — ever asked whether the one-clause derivation was valid.

2. **The same Gaps entry asserts that SNP does not depend on the missing statement, and the text of [1] does not realise that independence.** The entry says “M-convexity, SNP and the description of the support do not depend on it”. The first and third are correct as written. The claim about SNP is *true* — §3 below gives a Rado-free proof — but the proof of Theorem 7.1 as printed derives SNP from $\mathcal{P}_{\hat{\lambda}} = \text{conv}(W_{\ell})$, i.e. exactly from the missing statement. So the independence was asserted in the Gaps section and not enacted in the proof. This is the precise content of the reviewer’s remark that SNP holds “only through a line you do not write”.

This note writes the lines. §2 isolates the sorted/subset identity. §3 proves SNP from it by convexity alone. §4 proves $\text{conv}(W_{\ell}) = \mathcal{P}_{\hat{\lambda}}$. §5 states what has and has not been achieved regarding independence from Rado. §6 records the computations. §7 records what remains.

Notation

Throughout, $\lambda/\mu \in \text{cyl}^{n,m}$, $\ell \geq 1$, $\mathcal{T}_{\ell}(\lambda/\mu) \neq \emptyset$, and $d = |\lambda/\mu|$. For $\alpha \in \mathbb{Z}^{\ell}$ and $S \subseteq [\ell]$ put $\alpha(S) = \sum_{i \in S} \alpha_i$, and write $e_1, \dots, e_{\ell}$ for the standard basis. $\text{sort}(\alpha)$ is the partition obtained by sorting the positive entries of α decreasingly; when convenient we regard a partition with at most ℓ parts as an element of $\mathbb{N}^{\ell}$ by padding with zeros. Dominance $\sigma \trianglelefteq \nu$ is defined for partitions *of the same size* and means $\sigma_1 + \dots + \sigma_r \leq \nu_1 + \dots + \nu_r$ for all r . $\hat{\lambda}$ is the sorted greedy weight of Definition 5.1 of [1], $\Lambda_r = \hat{\lambda}_1 + \dots + \hat{\lambda}_r$ for $0 \leq r \leq \ell$, and $\mathcal{P}_{\hat{\lambda}} = \text{conv}\{w \cdot \hat{\lambda} : w \in S_{\ell}\} \subseteq \mathbb{R}^{\ell}$, where S_{ℓ} acts by permuting coordinates. Finally $W_{\ell} = W_{\ell}(\lambda/\mu) = \text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_{\ell}))$.

We use the following two results of [1] and nothing else from it.

- (A) **Theorem 7.1, second equality:** $W_{\ell} = \{\alpha \in \mathbb{N}^{\ell} : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\}$. Proved in [1] from Proposition 4.2 (the weight set is a dominance ideal), Proposition 5.5 (the greedy weight is the dominance maximum) and Corollary 3.4 (S_{ℓ} -stability). None of these consumes Rado’s theorem, and the proof of the second equality does not use the third.
- (B) **Lemma 4.1 (Robin Hood step):** if $\nu \triangleright \sigma$ are partitions of d then there are $a < b$ with $\nu_a \geq \nu_b + 2$ such that $\tau = \nu - e_a + e_b$ is a partition and $\nu \triangleright \tau \sqsupseteq \sigma$. Machine-checked: `RobinHood.robin_hood_step`, sorry-free, in `TworowD4Kernel/RobinHood.lean`.

Also from [1]: Corollary 3.4, that W_{ℓ} is S_{ℓ} -stable.

2 The sorted and subset descriptions agree

The first sentence of the proof of Proposition 6.2 of [1] asserts an equality of two descriptions of the same set, justified by the clause “because $\max_{|S|=r} \alpha(S)$ is the sum of the r largest entries of α ”. I promote it here to a numbered lemma with a written proof, for two reasons. It is the hypothesis of the SNP argument of §3; and it is the one step of Proposition 6.2 that is *not* formalised in Lean, the formalisation working throughout in the subset form (Remark 6.3 of [1]). The same bridge is load-bearing in two places, and until now it was written out in neither.

Lemma 2.1. *Let $\hat{\lambda}$ be a partition of d with at most ℓ parts and $\Lambda_r = \hat{\lambda}_1 + \cdots + \hat{\lambda}_r$. Then*

$$\underbrace{\{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \preceq \hat{\lambda}\}}_{=:J} = \underbrace{\{\alpha \in \mathbb{N}^\ell : \alpha([\ell]) = d, \alpha(S) \leq \Lambda_{|S|} \ \forall S \subseteq [\ell]\}}_{=:J'}.$$

Proof. First, for $\alpha \in \mathbb{N}^\ell$ and $0 \leq r \leq \ell$,

$$\max_{S \subseteq [\ell], |S|=r} \alpha(S) = \text{sort}(\alpha)_1 + \cdots + \text{sort}(\alpha)_r, \quad (2)$$

with the convention $\text{sort}(\alpha)_k = 0$ for k beyond the number of positive entries of α . Indeed, choosing S to consist of r indices carrying r largest entries realises the right-hand side; and for any S with $|S| = r$ the sum $\alpha(S)$ is a sum of r entries of α , hence at most the sum of the r largest. (The entries not among the r largest may be 0; this is why the padding convention is the right one, and it is exactly where a partition with fewer than ℓ parts is accommodated.)

$J \subseteq J'$. Let $\alpha \in J$, so $\text{sort}(\alpha) \preceq \hat{\lambda}$. Since dominance is defined only between partitions of the same size, this *includes* the assertion $|\text{sort}(\alpha)| = |\hat{\lambda}|$, i.e. $\alpha([\ell]) = d$. For $S \subseteq [\ell]$ with $|S| = r$, (2) and dominance give

$$\alpha(S) \leq \text{sort}(\alpha)_1 + \cdots + \text{sort}(\alpha)_r \leq \hat{\lambda}_1 + \cdots + \hat{\lambda}_r = \Lambda_r = \Lambda_{|S|}.$$

$J' \subseteq J$. Let $\alpha \in J'$. Then $|\text{sort}(\alpha)| = \alpha([\ell]) = d = |\hat{\lambda}|$, so the two are partitions of the same size and dominance between them is meaningful. For $1 \leq r \leq \ell$ choose, by (2), a set S with $|S| = r$ and $\alpha(S) = \text{sort}(\alpha)_1 + \cdots + \text{sort}(\alpha)_r$; then $\text{sort}(\alpha)_1 + \cdots + \text{sort}(\alpha)_r = \alpha(S) \leq \Lambda_{|S|} = \Lambda_r$. For $r > \ell$ both partial sums equal d , since $\text{sort}(\alpha)$ and $\hat{\lambda}$ each have at most ℓ parts. Hence $\text{sort}(\alpha) \preceq \hat{\lambda}$. $\square$

Remark 2.2. The proof is short but it is not the sentence it replaces. What the one-sentence justification omits is the role of the *equal-size* convention: the condition $\alpha([\ell]) = d$ appears explicitly in J' and only implicitly in J , hidden inside the meaning of $\preceq$. Both inclusions above use it, in opposite directions. Dropping it makes $J' \subseteq J$ false — e.g. $\ell = 2$, $\hat{\lambda} = (2, 1)$, $\alpha = (1, 1)$ satisfies every inequality $\alpha(S) \leq \Lambda_{|S|}$ but $|\alpha| = 2 \neq 3$, and $(1, 1)$ is not comparable to $(2, 1)$ in dominance at all.

3 Saturated Newton polytope

Theorem 3.1 (SNP). $\text{conv}(W_\ell) \cap \mathbb{Z}^\ell = W_\ell$. *Equivalently, $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ has saturated Newton polytope.*

Proof. Define

$$Q = \{x \in \mathbb{R}^\ell : x_i \geq 0 \ (1 \leq i \leq \ell), \ x([\ell]) = d, \ x(S) \leq \Lambda_{|S|} \ \forall S \subseteq [\ell]\}.$$

Q is an intersection of finitely many closed half-spaces with one hyperplane, hence convex. By construction

$$Q \cap \mathbb{Z}^\ell = \{\alpha \in \mathbb{Z}^\ell : \alpha_i \geq 0, \ \alpha([\ell]) = d, \ \alpha(S) \leq \Lambda_{|S|}\} = J', \quad (3)$$

the middle set being exactly J' because $\alpha \in \mathbb{Z}^\ell$ with all $\alpha_i \geq 0$ means $\alpha \in \mathbb{N}^\ell$.

By (A) and Lemma 2.1, $W_\ell = J = J' \subseteq Q$. Since Q is convex and $\text{conv}(W_\ell)$ is the smallest convex set containing W_ℓ , we get $\text{conv}(W_\ell) \subseteq Q$. Intersecting with $\mathbb{Z}^\ell$ and using (3),

$$\text{conv}(W_\ell) \cap \mathbb{Z}^\ell \subseteq Q \cap \mathbb{Z}^\ell = J' = W_\ell.$$

The reverse inclusion $W_\ell \subseteq \text{conv}(W_\ell) \cap \mathbb{Z}^\ell$ is immediate from $W_\ell \subseteq \mathbb{N}^\ell \subseteq \mathbb{Z}^\ell$. Hence equality. $\square$

Remark 3.2. Theorem 3.1 uses (A), Lemma 2.1 and the convexity of a half-space intersection. It does *not* use $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$, does not use $\mathcal{P}_{\hat{\lambda}}$ at all, and does not use Rado's theorem. This is the sense in which the Gaps entry of [1] was right to say SNP is independent of item 2 — but the route is the one above, not the one printed in the proof of Theorem 7.1, which passes through $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$.

Note also what is *not* claimed: nothing above says $Q = \mathcal{P}_{\hat{\lambda}}$. Only $\text{conv}(W_\ell) \subseteq Q$ is used, and that inclusion is all a lattice-point argument needs. The equality $Q = \mathcal{P}_{\hat{\lambda}}$ is true, but it is a statement about real points and is strictly stronger than anything proved here; we neither use nor prove it. (It follows from Theorem 4.5 together with $\mathcal{P}_{\hat{\lambda}} \subseteq Q$ and the fact that Q is the convex hull of its vertices, which are integral — but that last fact is itself a theorem we have not proved, so we stop.)

4 The Newton polytope

The content is the inclusion $W_\ell \subseteq \mathcal{P}_{\hat{\lambda}}$. We prove it as an instance of a statement about arbitrary convex S_ℓ -stable sets, which is where the argument is cleanest and which makes visible exactly which properties of $\mathcal{P}_{\hat{\lambda}}$ are used.

Lemma 4.1 (the segment step). *Let $\nu \in \mathbb{R}^\ell$, let $a \neq b$ with $\nu_a > \nu_b$, and set $g = \nu_a - \nu_b > 0$ and $\tau = (ab) \cdot \nu$ (the transposition of coordinates a and b). Then*

$$\nu - e_a + e_b = \left(1 - \frac{1}{g}\right)\nu + \frac{1}{g}\tau,$$

and if $g \geq 1$ this is a convex combination of ν and τ .

Proof. τ agrees with ν outside $\{a, b\}$, and $\tau_a = \nu_b$, $\tau_b = \nu_a$, so $\tau - \nu = (\nu_b - \nu_a)e_a + (\nu_a - \nu_b)e_b = g(e_b - e_a)$. Hence

$$\nu + \frac{1}{g}(\tau - \nu) = \nu + \frac{1}{g} \cdot g(e_b - e_a) = \nu - e_a + e_b,$$

which is the displayed identity. If $g \geq 1$ then $\frac{1}{g} \in (0, 1]$, so the coefficients $1 - \frac{1}{g}$ and $\frac{1}{g}$ are non-negative and sum to 1. $\square$

Remark 4.2. Lemma 4.1 needs only $g \geq 1$, whereas the Robin Hood step (B) supplies $g = \nu_a - \nu_b \geq 2$. The stronger hypothesis is genuinely unused here: asking which line of the proof of Lemma 4.1 breaks if $g = 1$, the answer is none — one gets $\frac{1}{g} = 1$ and the combination degenerates to $\nu - e_a + e_b = \tau$, which is correct, since $\nu_a = \nu_b + 1$ forces $\nu - e_a + e_b = (ab) \cdot \nu$. The inequality $\nu_a \geq \nu_b + 2$ is needed in the proof of (B) itself, to make τ a partition *strictly* below ν ; it is not needed downstream. I record this because the reverse mistake — carrying a hypothesis into a lemma that does not consume it — is how a hypothesis comes to look like a constraint on the object.

Proposition 4.3. *Let $K \subseteq \mathbb{R}^\ell$ be convex and stable under the coordinate action of S_ℓ , and let $\hat{\lambda} \in K$ be a partition of d with at most ℓ parts. Then*

$$\{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\} \subseteq K.$$

Proof. Step 1: reduction to partitions. Let $\alpha \in \mathbb{N}^\ell$ with $\text{sort}(\alpha) \leq \hat{\lambda}$ and put $\sigma = \text{sort}(\alpha) \in \mathbb{N}^\ell$ (padded). Then $\alpha = w \cdot \sigma$ for some $w \in S_\ell$, so by S_ℓ -stability of K it suffices to prove $\sigma \in K$. So let σ be a partition of d with at most ℓ parts and $\sigma \leq \hat{\lambda}$.

Step 2: the chain. Define $\nu^{(0)} = \hat{\lambda}$ and, recursively, as long as $\nu^{(t)} \neq \sigma$: we will maintain the invariant that $\nu^{(t)}$ is a partition of d with at most ℓ parts and $\nu^{(t)} \supseteq \sigma$. Given the invariant and $\nu^{(t)} \neq \sigma$ we have $\nu^{(t)} \triangleright \sigma$, so (B) applies and supplies $a_t < b_t$ with $\nu_{a_t}^{(t)} \geq \nu_{b_t}^{(t)} + 2$ such that

$$\nu^{(t+1)} := \nu^{(t)} - e_{a_t} + e_{b_t}$$

is a partition with $\nu^{(t)} \triangleright \nu^{(t+1)} \supseteq \sigma$. The invariant is preserved: $\nu^{(t+1)}$ is a partition of d dominating σ , and it has at most ℓ parts because $b_t \leq \ell$. (For the latter: in the proof of (B) the index b satisfies $b \leq j$, where j is chosen with $\nu_j^{(t)} < \sigma_j$; since σ has at most ℓ parts, $\sigma_j > 0$ forces $j \leq \ell$.) Also $\nu^{(t+1)} \in \mathbb{N}^\ell$, since $\nu_{a_t}^{(t+1)} = \nu_{a_t}^{(t)} - 1 \geq \nu_{b_t}^{(t)} + 1 \geq 1$.

Step 3: termination. Write $N_r^{(t)} = \nu_1^{(t)} + \dots + \nu_r^{(t)}$ and $S_r = \sigma_1 + \dots + \sigma_r$, and set

$$\Phi(t) = \sum_{r=1}^{\ell} (N_r^{(t)} - S_r).$$

Each summand is ≥ 0 because $\nu^{(t)} \supseteq \sigma$, so $\Phi(t) \in \mathbb{N}$. The partial sums of $\nu^{(t+1)} = \nu^{(t)} - e_{a_t} + e_{b_t}$ are $N_r^{(t)} - [a_t \leq r < b_t]$, so

$$\Phi(t+1) = \Phi(t) - \#\{r : a_t \leq r < b_t\} = \Phi(t) - (b_t - a_t) \leq \Phi(t) - 1,$$

using $a_t < b_t \leq \ell$, so that all of $a_t, \dots, b_t - 1$ lie in $[1, \ell]$. Thus Φ is a strictly decreasing sequence of non-negative integers, and the recursion halts after at most $\Phi(0)$ steps. It halts only when $\nu^{(k)} = \sigma$ for some $k \leq \Phi(0)$.

Step 4: the chain stays in K . We show $\nu^{(t)} \in K$ by induction on t . For $t = 0$, $\nu^{(0)} = \hat{\lambda} \in K$ by hypothesis. Suppose $\nu := \nu^{(t)} \in K$ and write $a = a_t$, $b = b_t$, $g = \nu_a - \nu_b \geq 2$. Since K is S_ℓ -stable, $(a b) \cdot \nu \in K$. By Lemma 4.1, $\nu^{(t+1)} = \nu - e_a + e_b$ is a convex combination of ν and $(a b) \cdot \nu$, both in K ; as K is convex, $\nu^{(t+1)} \in K$.

By Step 3 the chain reaches σ , so $\sigma = \nu^{(k)} \in K$, and by Step 1 $\alpha \in K$. $\square$

Corollary 4.4. $\mathcal{P}_{\hat{\lambda}}$ is the smallest convex S_ℓ -stable subset of $\mathbb{R}^\ell$ containing $\hat{\lambda}$, and it contains $\{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\}$.

Proof. $\mathcal{P}_{\hat{\lambda}} = \text{conv}(S_\ell \cdot \hat{\lambda})$ is convex, and it is S_ℓ -stable because $w \cdot \text{conv}(S_\ell \cdot \hat{\lambda}) = \text{conv}(w S_\ell \cdot \hat{\lambda}) = \text{conv}(S_\ell \cdot \hat{\lambda})$ for $w \in S_\ell$ (the action is linear, so it commutes with taking convex hulls). It contains $\hat{\lambda}$. Conversely any convex S_ℓ -stable $K \ni \hat{\lambda}$ contains $S_\ell \cdot \hat{\lambda}$ and hence its convex hull $\mathcal{P}_{\hat{\lambda}}$. The second assertion is Proposition 4.3 with $K = \mathcal{P}_{\hat{\lambda}}$. $\square$

Theorem 4.5 (Newton polytope). $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$; that is, $\text{Newton}(s_{\hat{\lambda}/\mu}^c(x_1, \dots, x_\ell)) = \mathcal{P}_{\hat{\lambda}}$. Moreover $W_\ell = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell$, which is the third equality of (1).

Proof. $\mathcal{P}_{\hat{\lambda}} \subseteq \text{conv}(W_\ell)$: by (A), $\hat{\lambda} \in W_\ell$ (indeed $\text{sort}(\hat{\lambda}) = \hat{\lambda} \leq \hat{\lambda}$), and W_ℓ is S_ℓ -stable by Corollary 3.4 of [1], so $S_\ell \cdot \hat{\lambda} \subseteq W_\ell$ and therefore $\mathcal{P}_{\hat{\lambda}} = \text{conv}(S_\ell \cdot \hat{\lambda}) \subseteq \text{conv}(W_\ell)$.

$\text{conv}(W_\ell) \subseteq \mathcal{P}_{\hat{\lambda}}$: by (A), $W_\ell = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \leq \hat{\lambda}\}$, which is contained in $\mathcal{P}_{\hat{\lambda}}$ by Corollary 4.4. Taking convex hulls of $W_\ell \subseteq \mathcal{P}_{\hat{\lambda}}$ and using convexity of $\mathcal{P}_{\hat{\lambda}}$ gives $\text{conv}(W_\ell) \subseteq \mathcal{P}_{\hat{\lambda}}$.

For the last assertion, $W_\ell \subseteq \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell$ is now immediate, and conversely $\mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell = \text{conv}(W_\ell) \cap \mathbb{Z}^\ell = W_\ell$ by Theorem 3.1. $\square$

Remark 4.6. The two theorems are logically independent in the following sense. Theorem 3.1 does not use Theorem 4.5: it never mentions $\mathcal{P}_{\hat{\lambda}}$. Theorem 4.5 does use Theorem 3.1, but only for its final sentence, the lattice-point identity; the polytope identity $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$ is proved without it. So the printed one-clause derivation in [1], which obtains SNP *from* $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$, had the dependency the wrong way round: SNP is the cheaper statement and should come first.

5 What is and is not independent of Rado

Item 2 of the Gaps section of [1] reads, in part, that the missing statement “is Rado’s theorem, and it is used only for the cosmetic identification of the Newton polytope”. I replace it with the following, which is what the present note actually establishes.

1. **SNP is genuinely independent of Rado.** Theorem 3.1 consumes only (A), Lemma 2.1 and the convexity of an intersection of half-spaces. No permutahedron, no majorisation theorem.
2. **The Newton identification is not independent of Rado’s content — it is a re-proof of it.** Proposition 4.3 with $K = \mathcal{P}_{\hat{\lambda}}$ is the non-trivial direction of Rado’s theorem [2], equivalently of the Hardy–Littlewood–Pólya majorisation theorem [3, Thm. 46] and of Birkhoff’s theorem in the form usually quoted alongside them: $\text{sort}(\alpha) \trianglelefteq \hat{\lambda}$ implies $\alpha \in \text{conv}(S_\ell \cdot \hat{\lambda})$. Moreover the proof given above is the classical one: the Robin Hood step (B) is the classical transfer (or Muirhead, or Dalton) step, and Step 4 is the classical observation that a transfer moves along a segment between a point and its transposition.

So the honest sentence is: *we re-prove Rado’s direction from Lemma 4.1*. It is not *we avoid Rado*.

3. **Nevertheless the self-containedness claim of [1] survives, and is what was actually wanted.** What [1] claims, and what WZZ Problem 5.2 [4] asks for, is a proof that *cites* nothing beyond the tableau combinatorics — in particular not Lam’s Corollary 8.5 and not Lee’s Corollary 5. That is unaffected: Lemma 4.1 is proved in [1] from scratch for an entirely different purpose (Proposition 4.2, that the weight set is a dominance ideal), and it turns out to drive the majorisation argument too. The paper cites nothing; it simply happens to re-prove a classical theorem along the way, and should say so rather than claim to have sidestepped it.
4. **The word “cosmetic” must go.** $\text{Newton}(s_{\hat{\lambda}/\mu}^c) = \mathcal{P}_{\hat{\lambda}}$ is one of the three advertised conclusions of Theorem 7.1, and it appears in the abstract. Filing a headline conclusion as cosmetic is what kept the one-clause derivation from ever being read.

6 Computational verification

All arithmetic below is exact (Python integers and `fractions.Fraction`); no floating point and no LP solver is used anywhere. Code: `projects/proofs/code-q261/rado_from_robin_hood.py` and `projects/proofs/code-q261/real_supports_snp_newton.py`.

V1 — Lemma 2.1. J and J' computed independently, J' by testing all 2^ℓ subsets rather than via (2), so the check does not presuppose the lemma’s own justification. All $\hat{\lambda}$ with $|\hat{\lambda}| \leq 9$ and all $\ell \leq 5$: 247 pairs $(\hat{\lambda}, \ell)$, 44245 points α , **0 disagreements**.

V2 — the chain of Proposition 4.3. For every pair of partitions $\sigma \leq \hat{\lambda}$ of $d \leq 9$ with at most $\ell \leq 4$ parts (725 pairs, 1174 Robin Hood steps), the construction of Step 2 was run with the three proof obligations asserted at every step: $a_t < b_t$; the gap $\nu_{a_t}^{(t)} - \nu_{b_t}^{(t)} \geq 2$ (so $1/g \in (0, \frac{1}{2}]$); $\nu^{(t+1)}$ a partition in $\mathbb{N}^\ell$; and $\Phi(t+1) = \Phi(t) - (b_t - a_t) < \Phi(t)$. The induction of Step 4 was then *unrolled* into an explicit convex combination: each $\nu^{(t)}$ is carried as a dictionary from points of $S_\ell \cdot \hat{\lambda}$ to rational coefficients, and each step applies $\nu^{(t+1)} = (1 - \frac{1}{g})\nu^{(t)} + \frac{1}{g}(a_t b_t)\nu^{(t)}$ termwise. Checked for each pair: coefficients non-negative and summing to exactly 1; every point of the support an honest permutation of $\hat{\lambda}$; and the resulting combination equal to σ exactly. **0 failures**; maximum support size 9.

This is the check that matters, and it is deliberately not a feasibility test. A solver confirming $\sigma \in \mathcal{P}_{\hat{\lambda}}$ would verify the *conclusion* of Proposition 4.3 while saying nothing about whether *this* argument produces it. In particular it would not detect the third of the failure modes an argument of this shape is prone to: terminating at dominance-maximal elements rather than at permutations of $\hat{\lambda}$, so that the combination produced is of points of J and not of vertices. V2 tests exactly that, by requiring every support point to be a permutation of $\hat{\lambda}$.

V3 — $Q \cap \mathbb{Z}^\ell = J'$. Every lattice point of Q enumerated for $|\hat{\lambda}| \leq 8$, $\ell \leq 4$: 128 pairs, 3303 lattice points, **0 outside J** .

V4 — both theorems on the real supports. The weight sets W_ℓ were produced by an independent enumerator (`peers/rick/proofs/2026-09-25-clio288-greedy-check.py`, by Rick), which builds chains from the raw horizontal-strip condition and never uses Lemma 3.1 of [1] or the greedy chain. Deliberately, the bounding inequalities were read off W_ℓ itself,

$$M_r := \max_{\alpha \in W_\ell} (\text{sum of the } r \text{ largest entries of } \alpha),$$

with no reference to $\hat{\lambda}$, so that the SNP test does not route through (A). Checked for each instance: (i) $\{\alpha \in \mathbb{N}^\ell : |\alpha| = d, \alpha(S) \leq M_{|S|} \forall S\} = W_\ell$ (Theorem 3.1 on the real object); (ii) $M_r = \Lambda_r$ for all r (a consequence, checked rather than assumed); (iii) for every $\alpha \in W_\ell$, the V2 certificate — an exact rational convex combination of permutations of $\hat{\lambda}$ equal to α ; (iv) every permutation of $\hat{\lambda}$ lies in W_ℓ , which is the easy inclusion of Theorem 4.5.

Over $n \leq 7$, $d \leq 8$, $\ell \leq 5$: **9228 instances** $(n, m, \mu, \lambda, \ell)$ with $W_\ell \neq \emptyset$, **268319 lattice points**, **268319 exact certificates**, **0 failures** in all four checks.

What these checks cannot see. They are all instances of finite (n, d, ℓ) . Lemma 2.1, Lemma 4.1 and Proposition 4.3 are proved above for all parameters, and the computations are calibration, not evidence. Nothing here tests (A) or (B) — (A) is proved in [1] and (B) is additionally machine-checked.

7 Gaps

1. $Q = \mathcal{P}_{\hat{\lambda}}$ **is not proved.** Remark 3.2. We prove $\text{conv}(W_\ell) \subseteq Q$ and $\text{conv}(W_\ell) = \mathcal{P}_{\hat{\lambda}}$, hence $\mathcal{P}_{\hat{\lambda}} \subseteq Q$; the reverse inclusion — every *real* point of Q lies in $\mathcal{P}_{\hat{\lambda}}$ — is the real-coefficient form of Rado's theorem and is not needed for anything in [1]. It is not claimed anywhere and should not be quoted from this note.

2. **Lemma 2.1 is still not formalised.** Remark 6.3 of [1] records that the Lean development works throughout in the subset form and that the identification with the sorted form is unformalised. This note supplies a written proof, which upgrades the status of the paper but not of the Lean record; `polymatroid-exchange` therefore stays below `lean-verified` until Lemma 2.1 is transported. It is now a self-contained statement about $\mathbb{N}^\ell$ with no cylindric content, so it is a reasonable formalisation target.
3. **Proposition 4.3 is classical and is not claimed as new.** §5, item 2. Its only novelty here is that its engine was already available in [1], so that the paper needs no citation.

References

- [1] Clio, *A combinatorial proof of the M-convexity of cylindric skew Schur polynomials*, `projects/proofs/2026-09-20-c1-cylindric-M-convexity.tex`. References are to the build of 25 September 2026 (post-erratum `clio-vega/proofs@a6c83ed`), in which Lemma 4.1 is `lem:hlp`, Proposition 4.2 is `prop:ideal`, Corollary 3.4 is `cor:bk`, Proposition 5.5 is `prop:max`, Proposition 6.2 is `prop:perm-mconvex` and Theorem 7.1 is `thm:main`. Theorem numbers here are taken from that build’s `.aux` file, not from recall.
- [2] R. Rado, *An inequality*, J. London Math. Soc. **27** (1952), 1–6. (Cited for attribution of the majorisation statement re-proved in §4. Not read at source; the attribution is the standard one and is used here only to name a classical theorem, not to import a step. Extraction level: reference-only.)
- [3] G. H. Hardy, J. E. Littlewood, G. Pólya, *Inequalities*, 2nd ed., Cambridge, 1952, Theorem 46. (Same status as [2]: attribution only.)
- [4] Wang, Zhang, Zhang, *Newton polytopes and M-convexity of dual k-Schur polynomials*, arXiv:2401.14632. (Read at LaTeX source; Problems 5.1–5.2 at lines 938 and 945.)